

Q1) Given, $x = p + r$ $p \in W$ $\{ r \in W^\perp$
 for $y \in W$

$$\|x - y\| = \|(p - y) + r\| = \sqrt{(p - y)^2 + r^2}$$

$$\|x - y\| = f(y) = \sqrt{(p - y)^2 + r^2}$$

where is magnitude of $x - y$

and we can see that $f(y)$ is minimum at $p = y$.
 Hence, claim is true.

Q2) 1) No, in such cases tie-breaking conditions must be used

2) Yes, as after each iteration the clustering ~~will~~ change
 based on density.

3) Yes, two different choices can lead to different
 final representative vectors as in one case it is grouped
 with c_1 & in another with c_2 .

Q3) $A^T - \lambda A = 0$

for all diagonal elements $a_{ii} - \lambda a_{ii} = 0$

hence i) $\lambda = 1$, ii) $a_{ii} = 0$ & $\lambda \in \mathbb{R}$

i) $A^T - A = 0$

$A^T = A$

$\therefore A$ is symmetric

ii) for a_{ij} in A

$a_{ji} - \lambda a_{ij} = 0 \Rightarrow a_{ji} = \lambda a_{ij}$

$a_{ij} - \lambda a_{ji} = 0$

$a_{ij} = \lambda^2 a_{ij}$

$a_{ij}(1 - \lambda^2) = 0$

A is symmetric

$\lambda = \pm 1$

or A is null matrix

Q5)

$$u^T v = 0 \Rightarrow \sum u_i v_i = 0$$

Thus shows that $u^T v$ can be zero without $u \cdot v = 0$

$$\text{In } u \cdot v \Rightarrow u_i v_i = 0$$

This shows that atleast one value is 0.
So it is possible if $u \cdot v \neq 0$.

$$\text{Ex - } u = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad v = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$u^T v = [0] = 0$$

$$u \cdot v = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0$$

$$\text{Ex - } u = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$u^T v = 0$$

$$u \cdot v = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \neq 0$$

Hence they are not same.

